

MODELING OF MULTI-PHYSICS PHENOMENA - LECTURE 1

Numerical approximation of PDEs with the finite Element Method (FEM)

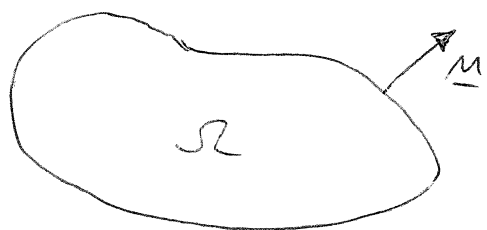
Poisson Equation

$$(1) \quad -\Delta u = f$$

$$u: \mathbb{R}^m \rightarrow \mathbb{R} \quad m=1,2,3$$
$$f: \mathbb{R}^m \rightarrow \mathbb{R} \quad \text{scalar functions}$$

u is the UNKNOWN, f is the DATA, and is given on a domain $\Omega \subseteq \mathbb{R}^m$

A solution u of (1) will also satisfy BOUNDARY CONDITIONS on the boundary $\partial\Omega$ of Ω :



$$(2) \quad \alpha u + \beta \frac{\partial u}{\partial n} = g \quad \text{on } \partial\Omega$$

$\underline{n}$ is normal vector on $\partial\Omega$, pointing outwards, g is data, $\frac{\partial u}{\partial n} = \nabla u \cdot \underline{n}$

Δu is the LAPLACIAN:

$$\Delta u = \sum_{i=1}^m \frac{\partial^2 u}{\partial x_i^2}$$

E.g. for $m=2$ (2D domain), $u = u(x, y)$ and $\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$

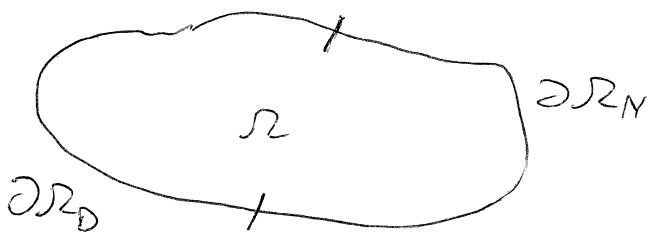
Combination of (1) and (2) gives a BOUNDARY VALUE PROBLEM:

$$\begin{cases} -\Delta u = f & \text{in } \Omega \\ \alpha u + \frac{\partial u}{\partial n} = g & \text{on } \partial\Omega \end{cases} \quad (\text{BVP})$$

Types of boundary conditions

- $\beta = 0, d = 1 \rightarrow u = g$ on $\partial\Omega$ DIRICHLET BC
- $\beta = 1, d = 0 \rightarrow \frac{\partial u}{\partial n} = g$ on $\partial\Omega$ NEUMANN BC
- $\beta \neq 0, d \neq 0 \rightarrow d u + \beta \frac{\partial u}{\partial n} = g$ on $\partial\Omega$ ROBIN BC

We can also have multiple BCs, each on a different part of $\partial\Omega$. Example:



$$u = g \quad \text{on } \partial\Omega_D$$

$$\frac{\partial u}{\partial n} = h \quad \text{on } \partial\Omega_N$$

WEAK FORMULATION

For simplicity now we consider the following 1D case:

$$(D) \quad \begin{cases} -u'' = f & \text{on } [0, 1] \\ u(0) = u(1) = 0 \end{cases}$$

homogeneous
Dirichlet BCs

$$\Omega = [0, 1]$$

To apply the FEM, we first rewrite the problem in an alternative formulation: instead of solving $-u''(x) - f(x) = 0$, we require that, for an appropriate set of TEST FUNCTIONS $v \in V$, it holds

$$\int_{\Omega} (-u'' - f) v \, dx = 0 \quad \forall v \in V \quad (3)$$

Notice that we haven't specified what the set V is. In equation (3) $\forall \psi \in V$ means that the equality must hold for any arbitrary test function, $\psi(x)$, which belongs to the set V . We can first integrate by parts in equation (3) to obtain

$$-\int_{\Omega} u'' \psi dx = \int_{\Omega} f \psi dx \Rightarrow -[u' \psi]_0^1 + \int_{\Omega} u' \psi' dx = \int_{\Omega} f \psi dx \quad \forall \psi \in V$$

$$\Rightarrow \int_{\Omega} u' \psi' dx = \int_{\Omega} f \psi dx \quad \forall \psi \in V \quad (w)$$

Equation (w) is the WEAK FORMULATION of the problem (D). Notice that the boundary term, $[u' \psi]_0^1$ equals to 0 due to boundary conditions. Solutions to (w) are called WEAK SOLUTIONS, while solutions to (D) are CLASSICAL SOLUTIONS. It is easy to see that any classical solution is a weak solution. The weak formulation exists as long as the integrals in (w) are well defined. Therefore, we choose the set V as the set of functions with enough regularity so that the integrals in (w) are computable. For our case, the correct choice is

$$V = \left\{ \psi: [0,1] \rightarrow \mathbb{R}, \psi(0)=0, \psi(1)=0, \psi \text{ is continuous, } \psi' \text{ is piece-wise continuous} \right\}$$

And we rewrite the problem as

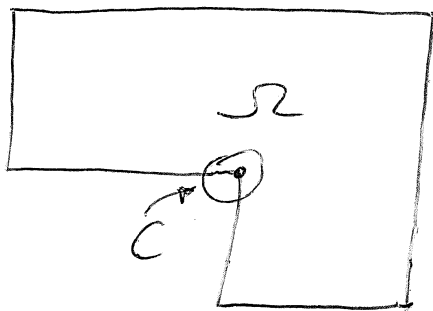
Find $u \in V$ such that, for any $v \in V$,

$$\int_{\Omega} u' v' dx = \int_{\Omega} f v dx$$

The weak formulation is the starting point of the Finite Element Method: we can use it to obtain a discrete approximation of the solution u .

Remarks on the weak formulation

Looking at the problem (1) we see that a classical solution has to be twice differentiable in Ω . On the other hand, the weak formulation has a weaker requirement on regularity: $u \in V$ needs only first derivative to be piece-wise continuous. In many engineering applications, a classical solution might not exist. For example:



Non-convex domain: the solution does not have a bounded second derivative at C

$$f = \begin{cases} 1 & \text{if } x \geq 0 \\ -1 & \text{if } x < 0 \end{cases}$$

discontinuous data prevents the solution from having continuous second derivative

Finite Element Method

$$(w) \quad \int_{\Omega} u' v' = \int_{\Omega} f v \quad \forall v \in V$$

The idea is to construct a DISCRETE SUBSET of V , called V_h , $V_h \subseteq V$, and to solve the weak formulation restricted on V_h .

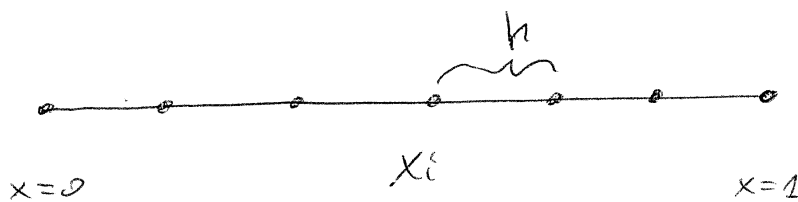
The space V_h will have a finite number of basis functions $\{\phi_1, \dots, \phi_N\}$ and

$$u_h \in V_h \Rightarrow u_h = \sum_{i=1}^N \sigma_i \phi_i \quad \sigma_i \in \mathbb{R}$$

linear combination

How do we build these functions?

First discretize the domain: divide it into SUB-INTERVALS. The endpoints of the sub-intervals are called NODES



$$x^i, \quad i=0, 1, \dots, N+1$$

$$x_0 = 0$$

$$x_{N+1} = 1$$

recall that $\Omega = [0, 1]$

$N+1$ intervals

$$\text{of length } h = \frac{1}{N+1}$$

On this discretized domain we construct a set of piece-wise polynomial functions, in particular piece-wise linear, called P1 functions

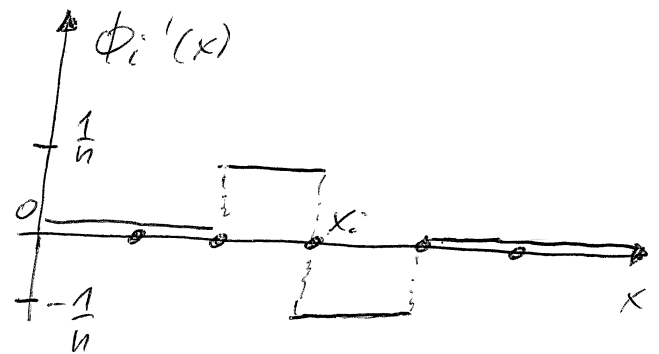
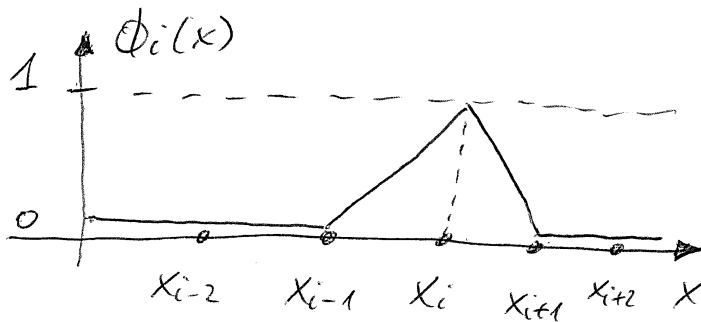
For each node x_i we have a function

$\phi_i(x)$ such that :

and also such that, restricted to each element, $\phi_i(x)$ is

Notice that we are a linear function.

Ignoring the nodes x_0 and x_{N+1} since the solution u and test functions v are 0 at the boundary due to the Dirichlet boundary conditions of the problem: each element of V is 0 at the boundary. Therefore there are N functions



ϕ_i is continuous, ϕ_i' piecewise-continuous

$\phi_i(0) = \phi_i(1) = 0$. Therefore $\phi_i \in V$

The set V_h is the set of functions that are linear combinations of the basis functions $\{\phi_1, \dots, \phi_N\}$, $V_h \subseteq V$

The derivations are easy to calculate

$$s_h \in V_h \Rightarrow s_h = \sum_{i=1}^N s_i \phi_i, \quad s_h' = \sum_{i=1}^N s_i \phi_i'$$

Now we define the discrete problem:

Find $u_h \in V_h$ such that

$$(5) \quad \int_{\Omega} u_h' s_h' dx = \int_{\Omega} s_h f dx \quad \forall s_h \in V_h$$

Substitute $u_h = \sum_{i=1}^N u_i \phi_i$, $u_h' = \sum_{i=1}^N u_i \phi_i'$, and

any $i = 1, \dots, N$ to get pick $s_h = \phi_i$ for

$$(4) \quad \int_{\Omega} \sum_{j=1}^N u_j \phi_j' \phi_i' dx = \int_{\Omega} \phi_i f dx \quad i=1, \dots, N$$

solving this for $s_h = \phi_i$, $i=1, \dots, N$ is the same as solving for any possible $s_h \in V_h$.

We can regroup the N equations from (4) in matrix form: define the matrix $\underline{A}$, vectors $\underline{u}$ and $\underline{f}$

$$A_{ij} = \int_{\Omega} \phi_j' \phi_i' dx$$

$$u_i = u_i$$

$$f_i = \int_{\Omega} \phi_i f dx$$

$$\underline{u} = \begin{bmatrix} u_1 \\ \vdots \\ u_N \end{bmatrix}$$

$$\underline{f} = \begin{bmatrix} f_1 \\ \vdots \\ f_N \end{bmatrix}$$

$$\underline{A} \underline{u} = \underline{f}$$

This is a linear algebraic problem, a solution can be found provided that the matrix $\underline{A}$ is invertible

$\underline{A}$ is called the STIFFNESS MATRIX.

The coefficients of the matrix $\underline{A}$ and of the vectors $\underline{f}$ depend on the functions ϕ_i , which are known by construction.

Therefore they can be calculated by numerical integration. The matrix $\underline{A}$ is clearly ~~symmetric~~ symmetric:

$$\underline{A}_{ij} = \int_{\Omega} \phi_i' \phi_j' dx = \int_{\Omega} \phi_j' \phi_i' dx = \underline{A}_{ji}$$

Due to the properties of the Poisson Equation, the matrix A is also POSITIVE DEFINITE.

Therefore, the problem $\underline{A} \underline{u} = \underline{f}$ has a unique solution which can be calculated with standard algebraic solvers.

Once the solution vector $\underline{u}$ is known, the solution u_h of (6) is simply

$$u_h = \sum_{i=1}^N u_i \phi_i, \quad u_i = \underline{u}_i$$

The question now is how is u_h related to the real solution u of the problem (w). To answer this we first need a way to measure how "close" two functions are.

Introduce the function norm

$$\| \sigma \| = \sqrt{\int_{\Omega} |\sigma|^2 dx}$$

$$\| \sigma \| \geq 0$$

$$\| \sigma \| = 0 \Rightarrow \sigma(x) \equiv 0$$

Now we have the following

Theorem If the solution u of the problem (W) has bounded second derivative u'' (that is, if u is "regular enough") then the discrete solution u_h of problem (G) is such that:

$$\|u' - u_h'\| \leq h \|u''\|$$

$$\|u - u_h\| \leq h^2 \|u''\|$$

where h is the length of the sub-intervals, $h = \frac{1}{N+1}$. If $h \rightarrow 0$, so do $\|u' - u_h'\|$ and

$\|u - u_h\|$. Notice that $\|u - u_h\| = 0$ implies ~~approx~~ that the two functions coincide almost everywhere.

Extra Remark we introduced the ϕ_i functions as P_1 , piece-wise linear polynomials.

One can construct piece-wise polynomials of higher order, for example the P_2

Functions are piece-wise quadratic

